ML_Workflow

1.Data Collection

Data collection is the first step in machine learning. It involves gathering relevant information from various sources. The quality of collected data directly affects model performance. For this project, student academic data was collected including study hours, attendance, assignments, and marks.

2. Data Cleaning

Data cleaning removes errors, duplicates, and missing values from the dataset. Clean data improves the accuracy of machine learning models. In this dataset, no missing values were present, so minimal cleaning was required**.**

3. Feature Selection

Feature selection identifies the most useful variables for prediction. Good features improve model accuracy and reduce complexity. Here, Study Hours, Attendance, and Assignments were selected as features.

4. Train-Test Split

The dataset is divided into training and testing sets. Training data teaches the model patterns while testing data evaluates performance. This project uses an 80%-20% split.

5. Model Training

The machine learning algorithm learns relationships between features and labels. Linear Regression finds the best-fit line that minimizes prediction error. The model is trained using the training dataset.

6. Testing

Testing evaluates the trained model using unseen data. This helps determine how well the model generalizes to new situations. Predictions are generated on the test dataset.

7. Evaluation

Evaluation measures model performance using metrics such as MAE and R^2 Score. These metrics indicate prediction accuracy and goodness of fit. Better values imply better performance.

8. Prediction

Prediction uses the trained model to estimate future outcomes. New feature values are provided to the model. The model then predicts expected marks.